2026 SPARK
Machine Learning Concepts

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MODULE 1

Introduction to Machine Learning Concepts

What is ML, types of learning, the ML pipeline, and ethical considerations in medical AI

What Is Machine Learning?

Machine learning is a subfield of AI in which algorithms learn patterns directly from data, rather than being explicitly programmed with rules.

Clinical Analogy

Like a physician developing diagnostic intuition after thousands of patients ; ML formalises pattern recognition mathematically, at unprecedented scale and consistency.

Why It Matters in Medicine

Modern hospitals generate terabytes of EHR, imaging, and genomic data daily. ML can detect patterns across millions of records in seconds ; far beyond human cognitive capacity.

Diagnostic support ; expert-level detection in radiology, pathology, dermatology

Risk stratification ; identifying high-risk patients for proactive, personalised care

Drug discovery ; predicting molecular interactions, reducing years of lab screening

Operational efficiency ; forecasting patient volumes, optimising OR schedules

Types of Machine Learning

Supervised Learning

Learns from labelled data ; the algorithm sees inputs AND correct answers.

Example: Predicting diabetes, classifying tumours, estimating length of stay

Unsupervised Learning

Finds hidden structure without labels ; discovers natural groupings on its own.

Example: Disease phenotyping, patient segmentation, anomaly detection

Reinforcement Learning

An agent learns to make sequential decisions by receiving feedback in the form of rewards or penalties after each action. Over many iterations it discovers the optimal strategy for a given environment.

Example: Optimising ventilator settings, personalising drug dosing

Semi-Supervised Learning

Combines small labelled + large unlabelled datasets for efficiency.

Example: Radiology AI with few expert-labelled scans + millions unlabelled

The Machine Learning Pipeline

Data Collection — Data Cleaning — Feature Engineering — Train/Test Split — Model Training — Evaluation — Deploy & Monitor

Data Collection

Sources: EHRs, imaging PACS, wearables, genomics. Must de-identify (HIPAA/PHIPA). Watch for selection bias ; who gets tested?

Data Cleaning

Handle missing values, outliers, inconsistent units. Clinical domain knowledge is irreplaceable ; only a clinician knows a glucose of 0 is impossible.

Train/Test Split

Prevent data leakage ; test set must be truly unseen. Split by patient, not encounter. In temporal data, train on past, test on future.

Evaluation & Deployment

Performance must be assessed using clinically meaningful metrics. For screening, sensitivity may be prioritised. For treatment decisions, specificity may matter more. Deployment requires integration with clinical workflows, clinician override mechanisms, and continuous monitoring for performance drift over time.

Ethical Considerations in Medical AI

Bias & Fairness

Models trained on one demographic may fail on others. Pulse oximetry algorithms overestimate O_2 saturation in darker skin. Always validate across subgroups before deployment.

Transparency & Interpretability

Clinicians need to understand WHY a model made a prediction. Linear/logistic regression are interpretable. Complex models may need SHAP or LIME for explanation.

Privacy & Security

Patient data must be de-identified and stored securely. Governed by HIPAA (US), PHIPA (Ontario), GDPR (EU), PIPEDA (Canada). Federated learning can train without sharing raw data.

Clinical Validation

External validation on independent cohorts is essential before deployment. Regulatory frameworks (Health Canada, FDA, EU MDR) are evolving for Software as a Medical Device (SaMD).

Preparing Data for Machine Learning

Data Splitting

The dataset is typically divided into 80% training and 20% testing. The training set is used to fit the model; the test set is held back for unbiased evaluation. Stratified splitting preserves the proportion of each class in both sets. The test set must never influence any preprocessing decisions such as scaling or feature selection.

Normalisation (Feature Scaling)

Clinical features are measured in different units with very different ranges. For example, insulin may range from 0 to 800 while BMI ranges from 18 to 60. Without scaling, large-range features dominate distance-based algorithms. Standardisation transforms each feature to mean 0 and standard deviation 1. The scaler must be fit on the training set only to prevent data leakage.

Feature Selection

Selecting the most relevant features reduces noise, improves model performance, and increases interpretability. The ANOVA F-score measures how well each feature discriminates between classes. Recursive Feature Elimination iteratively removes the least important features. Model-based approaches such as Random Forest feature importance rank variables by their contribution to prediction accuracy.

MODULE 2

Supervised Learning: Linear and Logistic Regression

Predicting continuous outcomes, classifying binary events, decision trees, random forests, and model evaluation

Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$$

Linear regression predicts a continuous numerical outcome by fitting a linear relationship between input features and the target variable. Each coefficient tells you how much the outcome changes per one-unit increase in that feature, holding all others constant. The algorithm finds the line (or hyperplane) that minimises the total sum of squared residuals using Ordinary Least Squares.

Metric	What It Measures	Clinical Meaning
MSE	Average of $(\text{actual} - \text{predicted})^2$	Penalises large errors heavily; units are squared
RMSE	$\sqrt{\text{MSE}}$	Error in original units ; "predictions off by X mmHg on average"
MAE	Average of $\lvert \text{actual} - \text{predicted} \rvert$	Average error magnitude, robust to outliers
R^2	1 − (unexplained / total variance)	Proportion of variance explained (0 = useless, 1 = perfect)

Key assumptions: linearity, independence, homoscedasticity, normal residuals

Linear regression is for continuous targets (e.g., blood pressure, length of stay). For binary outcomes (disease yes/no), use logistic regression ; linear regression predictions are not bounded to [0, 1] and cannot be interpreted as probabilities.

Logistic Regression

$$P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + ... + \beta_n x_n)})$$

The sigmoid function squeezes any value into [0, 1] ; output is a probability. Threshold (default 0.5) determines classification. Coefficients are interpretable as log-odds ratios.

Metric	Formula Intuition	Clinical Meaning
Accuracy	(TP+TN) / Total	Overall proportion correct ; misleading with imbalanced classes
Precision (PPV)	TP / (TP+FP)	"When the model says positive, how often is it right?"
Recall (Sensitivity)	TP / (TP+FN)	"Of all actual positives, how many does the model catch?"
F1-Score	Harmonic mean of precision & recall	Balances both error types ; useful with imbalanced data
AUC-ROC	Area under ROC curve	Overall discrimination: 0.5 = random, 0.8–0.9 = excellent, >0.9 = outstanding

Odds Ratios: e^b gives the odds ratio for each feature. OR > 1 = risk factor (increases odds per unit increase). OR < 1 = protective factor. OR = 1 = no effect. Directly interpretable in clinical terms.

The Confusion Matrix

The foundation of all classification metrics ; a 2×2 table of correct and incorrect predictions:

	Predicted: Negative	Predicted: Positive
Actual: Negative	**True Negative (TN)**	**False Positive (FP)**
Actual: Positive	**False Negative (FN)**	**True Positive (TP)**

False Negatives (FN): Missed cases ; the patient has the disease but the model says no. Most dangerous in screening.

False Positives (FP): False alarms ; the patient is healthy but the model says positive. Leads to unnecessary testing/treatment.

Precision vs. Recall ; A Clinical Decision

Prioritise RECALL (Sensitivity)

"Of all actual positives, how many does the model catch?"

When missing a case has severe consequences:

Cancer screening ; catch every cancer, even if some healthy people get further testing

Diabetes screening ; undiagnosed diabetes leads to retinopathy, nephropathy, neuropathy

Sepsis detection ; delayed treatment dramatically increases mortality

Infectious disease surveillance ; missing a case risks community spread

Prioritise PRECISION (PPV)

"When the model says positive, how often is it right?"

When false positives cause significant harm or cost:

Recommending surgery ; must be confident before invasive procedures

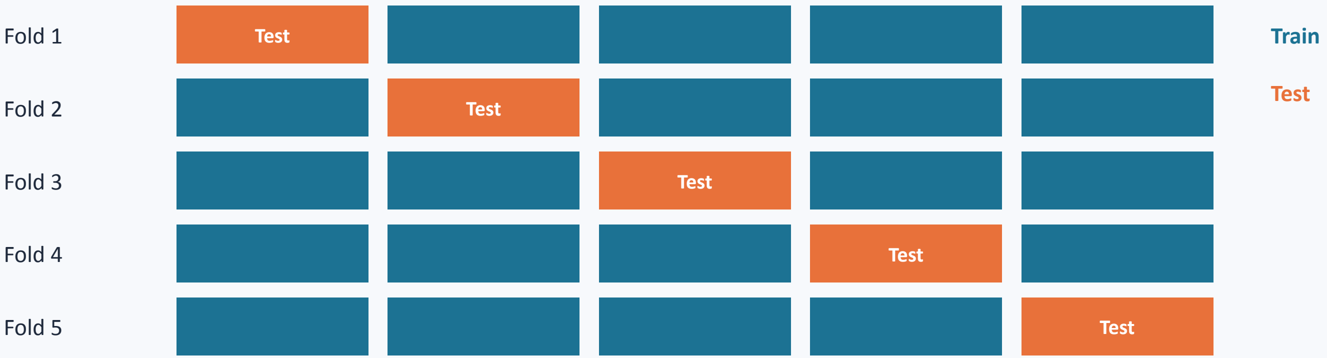
Prescribing toxic medications ; false positive = unnecessary toxicity exposure

Organ transplant matching ; false positive wastes a scarce organ

Rare disease diagnosis ; low prevalence means most positives are false without high precision

Cross-Validation

A single train/test split can produce optimistic or pessimistic results depending on which patients happen to fall in each set. K-Fold Cross-Validation addresses this by systematically rotating through multiple splits, providing a more robust and reliable estimate of true model performance.



How it works:

1. Divide data into K equal folds (typically K=5 or 10)
2. For each fold: use it as test set, train on remaining K−1 folds
3. Record performance for each fold
4. Report mean ± standard deviation across all folds

Stratified K-Fold preserves class proportions in each fold ; essential for imbalanced datasets.

Decision Trees & Random Forests

Decision Tree

A flowchart of yes/no questions:
"Is Glucose ≥ 127? → Is BMI ≥ 30? → Classify"

Splitting criteria:
Gini Impurity ; how often a random element would be misclassified (lower = purer)
Entropy ; information gain from a split (higher = better)

Advantages: interpretable, no scaling needed, handles non-linear relationships
Disadvantages: prone to overfitting, unstable (small data changes → different tree)

Random Forest

Ensemble of many diverse Decision Trees:

1. Bootstrap sampling ; each tree trained on random subset
2. Feature randomisation ; random subset of features per split
3. Majority vote ; final prediction from all trees combined

Advantages: reduces overfitting, more stable, handles noisy data, provides feature importance
Disadvantages: less interpretable, computationally heavier

Key hyperparameters: n_estimators (# trees), max_depth, max_features

MODULE 3

Unsupervised Learning: K-Means & Hierarchical Clustering

K-Means Clustering

1. Choose K (number of clusters)
2. Randomly initialise K centroids
3. Assign each patient to nearest centroid (Euclidean distance)
4. Recalculate centroids as the mean of assigned patients
5. Repeat steps 3–4 until convergence

Choosing K

Elbow Method: Plot WCSS vs K; look for the "elbow" where adding clusters gives diminishing returns

Silhouette Score:
Measures how well each point matches its cluster (−1 to 1; higher = better separation)

Domain knowledge:
The "best" K should also be clinically interpretable and actionable

Strengths & Limitations

✅ Fast, scalable, simple to implement
✅ Works well for roughly spherical, equal-sized clusters
❌ Must specify K in advance
❌ Sensitive to outliers and initial centroid placement
❌ Assumes spherical clusters ; struggles with irregular shapes

⚠️ Always standardise features before K-Means. Euclidean distance is dominated by large-range features otherwise.

Hierarchical (Agglomerative) Clustering

Builds a tree (dendrogram) showing how data points group at every level of similarity. **Unlike K-Means, you do NOT need to specify K in advance.**

Step 1

Start: each point is its own cluster

Step 2

Merge the two most similar clusters

Step 3

Repeat until one cluster remains

Step 4

Cut the dendrogram at chosen height → K clusters

Linkage Method	Merges Based On	Best For
Ward (most common)	Minimises within-cluster variance	Compact, equal-sized clusters
Complete	Maximum distance between clusters	Well-separated groups
Average	Mean pairwise distance	General purpose, robust to outliers
Single	Minimum distance between clusters	Elongated clusters (risk of chaining)

Advantages: No K needed upfront; dendrogram gives intuitive full hierarchy; doesn't assume spherical clusters

Disadvantages: Computationally expensive for large datasets ($O(n^2)$ memory); merges are irreversible; not practical for >10,000 observations

K-Means vs. Hierarchical ; When to Use Which

	K-Means	Hierarchical
Speed	Fast ; scales to millions	Slow ; practical for < 10,000
K required?	Yes ; must specify upfront	No ; cut dendrogram at any height
Cluster shape	Assumes spherical, equal-sized	Flexible ; any shape
Determinism	Varies by initialisation	Deterministic (same result every run)
Interpretability	Centroids are easy to profile	Dendrogram shows full hierarchy
Best for	Large datasets, quick exploration	Small–medium data, discovering hierarchy

Best practice: Run BOTH methods and compare using the Adjusted Rand Index (ARI). When both find similar groups, you gain confidence the structure is real and not an artefact of a particular algorithm. Moderate agreement is common and expected ; different algorithms define "similarity" differently.

Clinical applications: disease phenotyping (e.g., asthma endotypes, sepsis subtypes), patient stratification for targeted interventions, treatment response grouping, and hypothesis generation before confirmatory studies.

MODULE 4

Implementation of Machine Learning Methods

Pipelines, cross-validation, calibration, and clinical deployment considerations

Scikit-Learn Pipelines

A Pipeline chains all preprocessing and modelling steps into a single object. This is not just a convenience; it is essential for preventing data leakage, ensuring reproducibility, and simplifying deployment to production environments.

Imputer (median/mean) — **Scaler (standardise)** — **Model (classifier/regressor)**

Why use a Pipeline?

Prevents data leakage:

Scaler/imputer is fit on training data only, automatically ; no manual bookkeeping

Simplifies cross-validation: The entire pipeline is evaluated as one unit, ensuring preprocessing is done correctly within each fold

Eases deployment: One object handles everything from raw data to prediction ; serialize with joblib and deploy

Reproducibility: All steps are documented in a single, auditable object

Model Calibration

"When the model predicts 30% risk, do 30% of those patients actually have the disease?"

This is critical in medicine because predicted probabilities are communicated to patients for shared decision-making. A poorly calibrated model erodes clinical trust.

What is calibration?

A well-calibrated model's predicted probabilities closely match observed event frequencies.

How to assess it:

Plot predicted probability (x-axis) vs. observed frequency (y-axis). A perfectly calibrated model follows the diagonal.

Common patterns:

Curve above diagonal → model underestimates risk (dangerous in screening)

Curve below diagonal → model overestimates risk (causes unnecessary anxiety/testing)

Logistic regression is typically well-calibrated out of the box.

Complex models (Random Forest, neural networks) often need Platt scaling or isotonic regression for recalibration.

Clinical Deployment Checklist

Technical

- External validation on independent cohort
- Subgroup analysis across demographics
- Calibration assessment
- Sensitivity analysis for missing data
- Comparison vs. existing clinical tools

Clinical Integration

- Clear action pathway when model flags patient
- Clinician override mechanism
- Patient communication plan
- Accountability framework for decisions

Regulatory & Ethical

- Institutional ethics board / REB review
- Ongoing monitoring & retraining plan
- Algorithmic bias & fairness audit
- Patient informed consent process

Key Takeaways

Module 1

ML learns patterns from data at scale. Medical AI requires careful attention to bias, missing data, privacy, and clinical validation before deployment.

Module 2

Linear regression predicts continuous outcomes; logistic regression classifies binary events with interpretable odds ratios. Cross-validation provides robust performance estimates.

Module 3

K-Means partitions data into K groups; hierarchical clustering builds a dendrogram. Run both and compare ; validate clusters with clinical domain knowledge.

Module 4

Use Pipelines to prevent data leakage. Assess calibration for clinical trust. Plan thoroughly for deployment, monitoring, and regulatory compliance.